

Repurposed drugs for aneuploidy-induced stress

"BUB1B loss-of-function in mosaic variegated aneuploidy: which repurposed drugs target aneuploidy-induced proteotoxic stress? SAC deficiency causes chromosomal missegregation, aneuploidy, and pediatric rhabdomyosarcoma. Evaluate HSP90, WEE1, IGF-1R inhibitors, and senolytics. Rank by evidence strength and feasibility. Flag pediatric toxicity, contraindications for chromosomal instability, and missing data." *Generated by LeapSpace*

Repurposed Drugs for Aneuploidy-Induced Proteotoxic Stress in BUB1B Loss-of-Function MVA: Quick Insights

Key Findings Table

Drug class	Main rationale in BUB1B/MVA	Evidence strength	Feasibility in pediatrics	Key toxicity / CIN caution	Bottom line
WEE1 inhibitors	Best translational fit for chromosomal instability/replication-stress vulnerability rather than direct proteotoxic stress targeting; adavosertib has pediatric and adult activity, especially in combination regimens and TP53/DDR-vulnerable settings [1] [2] [3] [4] [5]	High for oncology translation; indirect for proteotoxic stress [1] [2] [3]	Moderate : pediatric experience exists, but toxicity is meaningful [3] [6]	Myelosuppression, diarrhea; caution with DNA-damage-amplifying combinations in developmentally vulnerable children; no MVA-specific safety data [6] [7]	Rank #1 overall for actionability, but not the best pure proteostasis strategy [3] [4]
HSP90 inhibitors	Best mechanistic match to aneuploidy-induced proteotoxic stress because aneuploid cells show impaired HSF1/HSP90 buffering; HSP90 blockade destabilizes multiple survival clients including AKT and IGF1R [8] [9] [10] [11]	Moderate overall; strong mechanistic alignment , weak direct BUB1B/MVA evidence [8] [9] [12]	Low-to-moderate : pediatric trials showed little objective activity [13] [14] [15]	Hepatotoxicity, ocular toxicity, QTc prolongation, diarrhea [16] [17] [18] [19] [20]	Rank #2 overall , but #1 mechanistically for proteotoxic stress [8] [9] [13]
IGF-1R inhibitors	Relevant to pediatric rhabdomyosarcoma lineage biology, but not to an established BUB1B/MVA-specific aneuploidy dependency [12] [21] [22]	Moderate for RMS pathway relevance; low for aneuploidy-directed repurposing [12] [21] [22]	Moderate tolerability, but limited efficacy [21] [22] [23]	Metabolic, hematologic, GI, constitutional toxicities; endocrine/growth risk remains insufficiently characterized in the provided pediatric evidence [21] [22] [23] [24]	Rank #3 overall ; weak fit for proteotoxic stress targeting [25] [26]

Drug class	Main rationale in BUB1B/MVA	Evidence strength	Feasibility in pediatrics	Key toxicity / CIN caution	Bottom line
Senolytics	Conceptually relevant only if SAC deficiency produces a bona fide senescent/SASP-high state; senescence in aneuploidy is heterogeneous and p53/context dependent [27] [28] [29] [30] [31] [32]	Low for BUB1B/MVA due to absence of direct testing [28] [33] [34]	Low in children [7] [35]	Navitoclax-class thrombocytopenia, neutropenia; dasatinib-associated hematologic/cardiac/QT risks [36] [37] [38] [39] [40]	Rank #4 overall; consider only after biomarker confirmation of senescence [29] [30] [41]

Direct Answer

Overall ranking by evidence strength and feasibility for BUB1B loss-of-function / mosaic variegated aneuploidy (MVA):

1) WEE1 inhibitors, 2) HSP90 inhibitors, 3) IGF-1R inhibitors, 4) senolytics [3] [4] [13] [21] [22] [34] [36].

If the question is narrowed to **which repurposed drugs most directly target aneuploidy-induced proteotoxic stress**, the mechanistic order changes to:

1) HSP90 inhibitors, 2) senolytics only in biomarker-confirmed senescent/SASP-high tumors, 3) WEE1 inhibitors, 4) IGF-1R inhibitors [1] [8] [9] [12] [29] [31].

The best **actionable** option today is **WEE1 inhibition**, because adavosertib has real pediatric and adult clinical experience and repeated combination activity, especially where replication stress or TP53/DDR vulnerability is present [1] [2] [3] [4] [5]. But that is an **indirect CIN strategy**, not a direct exploitation of proteostasis collapse [1] [42].

The best **mechanistic proteostasis strategy** is **HSP90 inhibition**, since aneuploid cells have impaired HSF1/HSP90 buffering and HSP90 inhibitors collapse multiple client pathways, including AKT and IGF1R [8] [9] [10] [11]. The limitation is translational: pediatric trials showed **minimal objective responses** and meaningful liver, ocular, and QTc toxicity concerns [13] [14] [16] [17] [18].

IGF-1R inhibitors are not strong aneuploidy-directed repurposing candidates. They have pediatric rhabdomyosarcoma relevance, but single-agent efficacy was poor and no retrieved evidence links BUB1B/SAC deficiency to IGF-1R dependence [12] [21] [22].

Senolytics remain the most speculative option. Aneuploidy can induce senescence and cGAS-STING/SASP signaling, but SAC deficiency does not uniformly do so, and no direct BUB1B/MVA or pediatric RMS senolytic studies were retrieved [27] [28] [30] [31] [34]. Navitoclax-class thrombocytopenia is a major pediatric barrier [35] [36] [38].

Confidence level: High - the ranking is strongly supported by the provided synthesis and concordant source patterns, though direct BUB1B/MVA-specific drug-testing evidence is absent [3] [12] [13] [28].

Study Scope

This review covers mechanistic aneuploidy biology, SAC/BUB1B deficiency, pediatric rhabdomyosarcoma and related solid-tumor drug studies, and early translational oncology evidence spanning foundational model systems, preclinical studies, and pediatric/adult clinical trials represented in the provided evidence set [3] [13] [21] [34] [43] [44]. Methods represented include mouse models, human cell lines, xenografts, pediatric phase I studies, adult combination trials, and reviews/meta-synthesis across these domains [1] [10] [14] [45] [46] [47].

Assumptions & Limitations

- No retrieved study directly tested **HSP90, WEE1, IGF-1R inhibitors, or senolytics in BUB1B-loss MVA models or patient-derived cells** [12] [28].
- No retrieved study directly measured **proteotoxic stress biomarkers** such as HSF1 activity, HSP90 client dependence, UPR burden, or proteasome addiction in BUB1B-deficient/MVA cells [12] [45].
- No retrieved pediatric RMS study linked response to **BUB1B status, aneuploidy burden, p53 status, or senescence phenotype** [12] [48].
- Syndrome-specific toxicity data for children with constitutional chromosomal instability are missing across all four classes [7] [23] [35].

Suggested Further Research

- Build **BUB1B-loss fibroblast, organoid, and RMS model panels** stratified by p53 status and aneuploidy burden, then compare HSP90 versus WEE1 versus senolytic sensitivity [5] [12] [27].
- Measure **HSF1/Hsp70 induction, UPR activation, autophagic flux, and proteasome dependence** directly in BUB1B-deficient models to validate the proteotoxic-stress hypothesis [8] [45] [49].
- Test whether **cGAS-STING/SASP biomarkers** identify a senolytic-responsive subset after SAC disruption or therapy-induced senescence [31] [32] [50].
- Define whether **WEE1 combinations** cause excess toxicity in nonmalignant tissues from constitutional CIN backgrounds [6] [7].
- In RMS, compare **HSP90 inhibition versus IGF-1R inhibition** in BUB1B-relevant models, since IGF-1R is an HSP90 client and HSP90 blockade may subsume narrower IGF-1R targeting [10] [51].

Full Report

Clinical and Biological Context

Foundational Context

BUB1B encodes BubR1, a core spindle assembly checkpoint (SAC) component. Loss or mutation impairs mitotic checkpoint control, causing chromosome missegregation, aneuploidy, developmental abnormalities, and cancer predisposition characteristic of mosaic variegated aneuploidy [52] [53] [54] [55]. In mouse and human systems, checkpoint failure is linked to chromosomal instability (CIN), and perturbation of SAC genes such as BubR1, Bub1, and Mad2 can promote aneuploidy and tumor formation, although the tumor effect is context dependent rather than universally oncogenic [56] [57] [58] [59] [60].

This matters therapeutically because aneuploidy is not just a karyotypic abnormality; it also creates a reproducible cellular stress state. Whole-chromosome imbalance increases sensitivity to protein synthesis and folding stress, alters stress-response programs, and increases reliance on protein-quality-control systems including proteostasis and proteasome-related functions [44] [61] [62] [63]. That creates a plausible rationale for repurposed drugs that exploit stress-handling dependencies rather than a lineage oncogene alone [9] [44].

At the same time, BUB1B deficiency can drive p53 activation, apoptosis, premature senescence, DNA damage, and inflammatory signaling, but these outcomes vary by model and genetic context [27] [28] [64] [65]. That heterogeneity is central to treatment choice: some strategies require a senescent state, whereas others require active proliferation and replication stress [29] [30] [31].

Confidence level: High - the SAC-to-aneuploidy-to-stress framework is directly supported by multiple foundational studies [44] [56] [62].

Genotype, phenotype, and cancer predisposition

BubR1 insufficiency in mice causes progressive aneuploidy, progeroid features, infertility, and senescence phenotypes, showing that reduced BubR1 affects both genome maintenance and organismal aging-like biology [43]. In neural tissue, BubR1 deficiency triggers chromosomal segregation defects, DNA damage, p53 activation, and apoptosis, but only partial rescue by Trp53 co-deletion indicates that non-p53 death pathways also operate [64]. Lower BubR1 levels can also induce p53/p21-dependent senescence, while p53 loss permits highly aneuploid proliferation [27]. Similar p53 dependence is seen after MAD2 depletion in fibroblasts: cells with intact p53/p21 undergo premature senescence, whereas INK4A/ARF-deficient backgrounds tolerate aneuploidy without strong senescence [28].

Taken together, these findings suggest that BUB1B/MVA tumors may not share a single dominant druggable state. Some may be proteostasis-stressed and proliferative; others may be damaged and senescent; still others may bypass arrest if p53 pathways are compromised [27] [28] [29]. That is why a biomarker-led rather than class-led strategy is more defensible [50] [65].

Confidence level: High - the evidence for heterogeneous outcomes after SAC failure is consistent across several systems [27] [28] [30] [64].

Why Aneuploid Cells Are Drug-Sensitive

Proteotoxic stress and chaperone dependence

Aneuploid human cells show impaired HSF1 and HSP90 function, and HSF1 is described as a key factor mitigating aneuploidy-associated proteotoxic stress [8]. Independently, aneuploidy is associated with sensitivity to protein-folding stress and increased dependence on proteostasis and ubiquitin-proteasome functions [44] [61] [62] [63]. HSP90 itself stabilizes client proteins, and its inhibition redirects destabilized clients toward proteasomal degradation [66] [67].

Broader stress pathways reinforce this logic. ER stress activates the UPR via PERK, IRE1, and ATF6, and the response can shift from adaptive to apoptotic depending on stress severity and duration [49] [68] [69] [70] [71] [72]. Autophagy is often a protective arm of this response, although it may fail or become suppressed as stress progresses [49] [72] [73] [74]. Interference with mitochondrial HSP90 can also activate AMPK, inhibit mTORC1, induce autophagy, and trigger ER UPR, illustrating how chaperone failure propagates into multi-system stress programs [75].

The key limitation is translational specificity: no retrieved paper directly measured these proteostasis variables in BUB1B-deficient or MVA cells [12] [45]. So the therapeutic rationale is strong, but still extrapolative.

Confidence level: High for the general aneuploidy-proteostasis model; **Low-to-medium** for direct applicability to BUB1B/MVA because direct measurements are missing [8] [12].

Senescence, apoptosis, and inflammatory signaling

Aneuploidy and SAC disruption can produce senescence, but not uniformly. Reduced BubR1 and MAD2 depletion can induce p53/p21-linked senescence [27] [28], and whole-chromosome instability after BUB1 or SMC1A reduction is associated with DNA breaks, oxidative stress, and a SASP-like phenotype [65]. Yet BUB1B/BubR1 or Bub3 loss and Mps1 inhibition did not produce widespread senescence in another timeframe/model, indicating that SAC bypass alone may be insufficient [30].

Inflammatory signaling provides a second layer of vulnerability. Micronuclei and cytosolic DNA can activate cGAS-STING, which drives interferon and SASP programs [31] [32] [50] [76] [77]. Persistent DNA damage signaling through ATM, NBS1, and CHK2 can sustain inflammatory cytokine secretion including IL-6 without depending on p53 or pRb for the secretory output itself [78]. This supports the idea that some highly unstable cells may transition into a senescent, inflammatory, potentially senolytic-sensitive state [31] [79].

However, no retrieved study shows that SASP intensity, cGAS-STING activation, or senescence markers predict drug sensitivity in BUB1B-deficient cells [29] [30] [50]. Therefore senolytic use without biomarkers would be empiric rather than evidence-based.

Confidence level: **High** for heterogeneity of senescence biology; **Low** for using it as a selection biomarker in BUB1B/MVA today [29] [30].

HSP90 Inhibitors

Mechanistic rationale and preclinical activity

HSP90 inhibitors are the most directly aligned with the proteotoxic-stress hypothesis. Aneuploid cells have impaired HSF1/HSP90 buffering [8] [9], and HSP90 inhibitors destabilize multiple oncogenic client proteins at once, including AKT and IGF1R, while inducing client degradation, apoptosis, or G2/M arrest across tumor models [10] [11] [80] [81] [82] [83] [84] [85]. This multi-client effect is especially attractive in genetically plastic CIN tumors, where single-pathway inhibition is often bypassed [11] [81] [83].

Preclinical pediatric solid-tumor work showed HSP90 inhibitors could deplete AKT and IGF1R and combine greater-than-additively with cisplatin [10]. Other combinations, such as PU-H71 plus bortezomib in Ewing sarcoma models, further support the idea that proteostasis-directed combinations may outperform monotherapy [86].

A useful inference from the provided data is that HSP90 inhibition may partly subsume IGF-1R targeting, because IGF1R is among the affected client pathways [10] [51]. **Inferred:** this makes standalone IGF-1R blockade less compelling in BUB1B-related disease when the aim is broad stress-network collapse rather than one receptor pathway. **Confidence level:** **Low** - this is a cross-source inference, not a direct comparative trial result [10] [51].

Pediatric rhabdomyosarcoma and solid-tumor data

The main translational weakness is efficacy. Pediatric phase I studies of 17-AAG in recurrent/refractory solid tumors reported **no objective responses**; one trial identified an MTD of 270 mg/m² with DLTs including grade 3 transaminitis and hypoxia, while another recommended 360 mg/m²/day without DLTs in the evaluated schedule [13] [14]. Pharmacodynamic evidence confirmed target engagement, including Hsp70/Hsp72 induction and assessment of Akt changes [13] [14]. In pediatric preclinical testing, AT13387 showed only modest single-agent activity and no objective xenograft responses [15].

Thus, HSP90 inhibition has **mechanistic strength but limited pediatric proof of efficacy** [13] [15]. The better near-term use case appears to be rational combinations rather than monotherapy [10] [86].

Toxicity and feasibility in children

Toxicity is a major constraint. HSP90 inhibitors have documented hepatotoxicity/liver enzyme elevation, ocular toxicity including retinal dysfunction and blurred vision, QTc prolongation, and common diarrhea [16] [17] [18] [19] [20]. Pediatric pharmacology reviews also emphasize that children's developing organs increase ADR vulnerability and justify long-term follow-up for growth and organ effects [7].

For a constitutional CIN syndrome such as MVA, the provided evidence does **not** prove excess syndrome-specific toxicity, but it equally does not exclude it [7]. That uncertainty lowers feasibility.

Confidence level: High for mechanism and toxicity profile; **Medium** for translational ranking because efficacy data are limited and indirect for BUB1B/MVA [8] [13] [17].

WEE1 Inhibitors

Mechanistic rationale in chromosomal instability

WEE1 inhibition does not directly attack proteotoxic stress. Instead, it amplifies replication stress, DNA damage, checkpoint failure, and mitotic catastrophe, which can be useful in CIN-high or TP53/DDR-vulnerable tumors [1] [2] [5] [42] [87]. In practical terms, this makes WEE1 a strong strategy for the *consequences* of SAC deficiency rather than the proteostasis defect itself [1] [42].

Adult studies support this mechanism repeatedly. Adavosertib showed activity with carboplatin in platinum-refractory ovarian cancer and demonstrated target engagement in multiple DNA-damage combination settings [1] [2] [88] [89]. Preclinical work also found synergy with 5-FU, cytarabine, and HDAC inhibition, with increased DNA damage and apoptosis [42] [87] [90]. PTEN loss and DNA-PK-related vulnerability were linked to greater sensitivity in one breast cancer study, showing that biomarker-defined subsets may matter [5].

Pediatric and adult clinical activity

Among the four classes evaluated, WEE1 has the strongest pediatric translational signal. Pediatric trials of adavosertib with irinotecan reported objective responses or prolonged stable disease in relapsed neuroblastoma and other solid/CNS tumors [3] [4] [91]. Adult combination studies additionally support feasibility with gemcitabine-radiation and platinum regimens [1] [92].

This is why WEE1 ranks first overall: not because it is the best match to proteotoxic stress, but because it has the best existing bridge from mechanism to clinic [1] [3].

Safety limitations and missing CIN-specific data

The main cited toxicities are myelosuppression and diarrhea [6]. Pediatric safety data remain limited, and no retrieved study addressed constitutional CIN syndromes specifically [7]. No direct BUB1B/MVA model testing was found [12], and no provided study directly examined WEE1 plus ATR inhibition in this context [93].

So WEE1 is the most actionable option, but with an important caution: in a developmentally vulnerable child with constitutional chromosomal instability, DNA-damage-intensifying regimens may risk nonmalignant tissue injury, and the provided literature does not resolve that risk [6] [7].

Confidence level: High for ranking WEE1 first overall; **Medium** for applying it to MVA specifically because the evidence is indirect [1] [3] [12].

IGF-1R Inhibitors in Pediatric Rhabdomyosarcoma

Monotherapy outcomes

IGF-1R inhibitors are clinically familiar in pediatric sarcoma, but efficacy as monotherapy is poor. R1507 in recurrent/refractory RMS, osteosarcoma, and synovial sarcoma produced a 2.5% objective response rate and very short median PFS, while pediatric RG1507 studies reported no objective responses though some patients achieved stable disease beyond 12 weeks [21] [22]. In recurrent/refractory sarcoma, only one RMS partial response was reported, and some tumor shrinkage in RMS was transient and lasted under 4 weeks [21]. Reviews across pediatric sarcoma trials consistently describe only modest or disappointing activity [25] [26] [94].

Resistance mechanisms and combination logic

The weakness of IGF-1R monotherapy is biologically understandable in the provided evidence. Resistance can arise via RTK redundancy and bypass signaling, including IRS1/PI3K/STAT3/p38 MAPK upregulation and other compensatory pathways [95] [96] [97]. In RMS models, synergy has been reported with HER2 inhibition, Src-family/YES inhibition, PDGFR α blockade, and ALK inhibition [46] [98] [99] [100] [101].

However, none of these RMS studies explicitly modeled chromosomal instability or aneuploidy-driven biology [101]. That sharply limits their relevance to BUB1B/MVA-specific repurposing.

Pediatric safety and BUB1B-specific gaps

Pediatric and mixed-age studies describe IGF-1R antibodies as generally tolerable but with metabolic, hematologic, GI, and constitutional toxicities; combination cohorts reported diarrhea and neutropenia as DLTs in some settings [21] [22] [102]. Later reviews note lack of survival benefit alongside increased adverse events, but the provided text does not enumerate endocrine-growth toxicities in detail [23]. Crucially, no retrieved paper linked SAC deficiency, MVA, or BUB1B loss to altered IGF-1R/INSR/PI3K-AKT signaling [12] [48] [103] [104] [105].

Accordingly, IGF-1R inhibitors are better viewed as **RMS pathway agents** than as **aneuploidy-selective proteotoxic-stress drugs** [12] [21].

Confidence level: High for weak monotherapy efficacy and missing BUB1B rationale [12] [21] [22].

Senolytics and Senescence-Targeting Strategies

Senescence biology in aneuploidy and SAC deficiency

Aneuploidy can induce senescence and pro-inflammatory SASP via micronucleus-driven cGAS-STING activation [31] [32] [47]. SAC depletion studies also support senescence-like programs in some contexts [28] [33] [65]. Yet this state is not universal: senescence varies with p53 status and model system, and SAC bypass alone may fail to generate widespread senescence [29] [30]. This means senolytics are only biologically plausible in a subset.

Evidence for senolytic combinations

The “one-two punch” concept is supported in broader oncology: chemotherapy or radiotherapy induces senescence, followed by senolytics such as navitoclax, FOXO4-DRI, or dasatinib plus quercetin to remove senescent residual cells [34] [106] [107] [108] [109]. This is translationally interesting because therapy-induced senescent cells may contribute to recurrence, metastasis, and bystander effects through SASP [108] [109].

But the evidence gap is decisive: no retrieved study directly tested senolytics in BUB1B-deficient, SAC-deficient, pediatric solid-tumor, or rhabdomyosarcoma models [110] [111] [112]. So the approach is conceptually attractive but experimentally immature for this use case [34].

Toxicity and pediatric feasibility

Navitoclax and related BCL-XL-directed senolytics have clear platelet liability, with thrombocytopenia repeatedly described as dose-limiting; neutropenia is also reported [36] [37] [38] [40] [113] [114] [115]. Dasatinib carries hematologic dysfunction, fluid retention, QT prolongation, arrhythmia/palpitations, and rare ventricular dysfunction or heart failure [36] [39] [40]. Pediatric use is further limited by sparse safety data and concerns regarding vulnerable organs and developing immunity [7] [35].

New platelet-sparing BCL-XL degraders are being developed [38] [116] [117], but these are better viewed as future design solutions than present repurposing answers for MVA.

Confidence level: High that senolytics are the most speculative and least feasible current option [34] [35] [36].

Comparative Ranking and Translational Prioritization

Ranked summary

1. WEE1 inhibitors

Best supported by pediatric/adult clinical activity and combination feasibility, especially in replication-stress and TP53/DDR-vulnerable settings [1] [2] [3] [4]. Weakness: indirect relevance to proteotoxic stress [42].

2. HSP90 inhibitors

Best mechanistic fit for aneuploidy-induced proteotoxic stress because aneuploid cells are HSF1/HSP90-impaired and HSP90 blockade collapses multiple survival clients [8] [9] [10] [11]. Weakness: pediatric objective activity has been poor and toxicity is substantial [13] [16] [17].

3. IGF-1R inhibitors

Have disease familiarity in pediatric RMS and rational combinations, but monotherapy efficacy is poor and there is no BUB1B/MVA-specific dependency signal [12] [21] [22].

4. Senolytics

Require a biomarker-confirmed senescent/SASP-high state, which is not consistently produced by SAC deficiency, and pediatric toxicity is problematic [30] [31] [35] [36].

Pediatric contraindications and CIN-specific cautions

- **HSP90 inhibitors:** hepatotoxicity, ocular toxicity, QTc prolongation, diarrhea; use would require stringent liver, ophthalmic, and ECG monitoring [16] [17] [18].
- **WEE1 inhibitors:** myelosuppression and diarrhea; special caution for intensified DNA-damage combinations in developmentally vulnerable children [6] [7].
- **IGF-1R inhibitors:** metabolic, hematologic, GI, and constitutional adverse events are reported, but detailed pediatric endocrine-growth risk remains insufficiently characterized in the provided evidence [21] [22] [23].
- **Senolytics:** navitoclax-class thrombocytopenia and neutropenia are major barriers; dasatinib adds cardiac/QT and fluid-retention risk [36] [38] [39] [40].

For **constitutional chromosomal instability**, the strongest general caution is that no retrieved study defines syndrome-specific excess toxicity for any class [7] [23] [35]. Clinical extrapolation should therefore assume uncertainty, not safety.

Missing data that limit immediate use

The field is limited by a model mismatch: mechanistic aneuploidy studies largely use generic cell systems and mouse models, whereas drug studies come from broader oncology rather than BUB1B/MVA-specific disease [12] [28] [45] [118]. Missing elements include:

- Direct drug testing in **BUB1B-loss/MVA models** [12].
- Direct mapping of **proteostasis biomarkers** in those models [45].
- CIN-stratified or BUB1B-stratified pediatric RMS response data [48].
- Definition of when SAC-deficient tumors are truly **senolytic-eligible** [30] [65].
- Long-term pediatric safety in children with developmental vulnerability and constitutional CIN [7] [35].

Overall conclusion confidence: High - the comparative ranking is strongly grounded in the supplied evidence and consistent across mechanistic, pediatric, and clinical domains, but direct BUB1B/MVA validation is absent [3] [12] [13] [35].

Conclusion

For **BUB1B loss-of-function MVA**, the most defensible current translational choice is **WEE1 inhibition** if the immediate priority is clinical feasibility, particularly in combination regimens and replication-stress-prone tumors [1] [3]. If the priority is to target **aneuploidy-induced proteotoxic stress itself**, **HSP90 inhibition** is the strongest mechanistic option, but present pediatric efficacy and toxicity data are not yet persuasive enough for confident syndrome-directed use [8] [9] [13] [17]. **IGF-1R inhibitors** remain lineage-relevant but biologically unconvincing as an aneuploidy-directed strategy [12] [21]. **Senolytics** should currently be viewed as hypothesis-generating tools for biomarker-selected settings rather than practical repurposed therapy in pediatric MVA [30] [35].

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